

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKENDDEVELOPMENT

DBMS Transactions Part-2:

```
Microsoft Windows [Version 10.0.26200.8894]
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C:\Users\hp>cd "C:\Program Files\MySQL\MySQL Server 8.0\bin"

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 22
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE city_library;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'c
ity_library' at line 1
mysql> CREATE DATABASE city_library;
Query OK, 1 row affected (0.06 sec)

mysql> USE city_library;
Database changed
mysql> CREATE TABLE Departments(
-> dept_id INT PRIMARY KEY,
-> dept_name VARCHAR(50)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Employees(
-> emp_id INT PRIMARY KEY,
-> emp_name VARCHAR(100),
-> salary DECIMAL(10,2),
-> dept_id INT,
-> manager_id INT,
-> email VARCHAR(100),
-> FOREIGN KEY(dept_id) REFERENCES Departments(dept_id)
```

```
-> dept_id INT,
-> manager_id INT,
-> email VARCHAR(100),
-> FOREIGN KEY(dept_id) REFERENCES Departments(dept_id)
-> );
Query OK, 0 rows affected (0.14 sec)

mysql> CREATE TABLE Customers(
-> customer_id INT PRIMARY KEY,
-> customer_id INT PRIMARY KEY,
-> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ')'
' at line 4
mysql> CREATE TABLE Customers(
-> customer_id INT PRIMARY KEY,
-> customer_name VARCHAR(100)
-> );
Query OK, 0 rows affected (0.10 sec)

mysql> CREATE TABLE Orders(
-> order_id INT PRIMARY KEY,
-> customer_id INT,
-> order_date DATE,
-> FOREIGN KEY(customer_id) REFERENCES Customers(customer_id)
-> );
Query OK, 0 rows affected (0.12 sec)

mysql> CREATE TABLE Products(
-> product_id INT PRIMARY KEY,
-> product_name VARCHAR(100),
-> price DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE Order_Details(
-> order_id INT,
-> product_id INT,
-> quantity INT,
-> FOREIGN KEY(order_id) REFERENCES Orders(order_id),
-> FOREIGN KEY(product_id) REFERENCES Products(product_id)
-> );
```

```
-> FOREIGN KEY(product_id) REFERENCES Products(product_id)
-> );
Query OK, 0 rows affected (0.12 sec)
```

```
mysql> INSERT INTO Departments VALUES
```

```
-> (1,'CSE'),
-> (2,'ECE'),
-> (3,'MBA'),
-> (4,'Civil');
```

```
Query OK, 4 rows affected (0.08 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Employees VALUES
```

```
-> (101,'Rahul',50000,1,NULL,'rahul@gmail.com'),
-> (102,'Priya',65000,1,101,'priya@gmail.com'),
-> (103,'Amit',75000,2,101,'amit@gmail.com'),
-> (104,'Neha',28000,3,103,'neha@gmail.com'),
-> (105,'Rohan',90000,2,NULL,'rohan@gmail.com'),
-> (106,'Sneha',55000,NULL,105,'sneha@gmail.com');
```

```
Query OK, 6 rows affected (0.01 sec)
Records: 6 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Customers VALUES
```

```
-> (1,'Ravi'),
-> (2,'Anita'),
-> (3,'Kiran');
```

```
Query OK, 3 rows affected (0.05 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Orders VALUES
```

```
-> (1,1,'2026-01-10'),
-> (2,2,'2026-02-12');
```

```
Query OK, 2 rows affected (0.03 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Products VALUES
```

```
-> (1,'Laptop',60000),
-> (2,'Mouse',500),
-> (3,'Keyboard',1200);
```

```
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```

mysql> INSERT INTO Order_Details VALUES
-> (1,1,2),
-> (1,2,3),
-> (2,2,5),
-> (2,3,2);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Employees
-> VALUES(107,'Kavya',62000,1,101,'kavya@gmail.com');
Query OK, 1 row affected (0.01 sec)

mysql> UPDATE Employees
-> SET salary = salary*1.10
-> WHERE dept_id=
-> (
-> SELECT dept_id
-> FROM Departments
-> WHERE dept_name='CSE'
-> );
Query OK, 3 rows affected (0.09 sec)
Rows matched: 3 Changed: 3 Warnings: 0

mysql> DELETE FROM Employees
-> WHERE salary<30000;
Query OK, 1 row affected (0.07 sec)

```

```

mysql> SELECT * FROM Employees;
+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | salary | dept_id | manager_id | email |
+-----+-----+-----+-----+-----+-----+
| 101 | Rahul | 55000.00 | 1 | NULL | rahul@gmail.com |
| 102 | Priya | 71500.00 | 1 | 101 | priya@gmail.com |
| 103 | Amit | 75000.00 | 2 | 101 | amit@gmail.com |
| 105 | Rohan | 90000.00 | 2 | NULL | rohan@gmail.com |
| 106 | Sneha | 55000.00 | NULL | 105 | sneha@gmail.com |
| 107 | Kavya | 68200.00 | 1 | 101 | kavya@gmail.com |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.05 sec)

```

```

mysql> SELECT emp_name,dept_name
-> FROM Employees
-> JOIN Departments
-> ON Employees.dept_id=Departments.dept_id;
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Rahul | CSE |
| Priya | CSE |
| Kavya | CSE |
| Amit | ECE |
| Rohan | ECE |
+-----+-----+
5 rows in set (0.00 sec)

```

```

mysql> SELECT *
-> FROM Employees
-> LEFT JOIN Departments
-> ON Employees.dept_id=Departments.dept_id
-> WHERE Departments.dept_id IS NULL;
+-----+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | salary | dept_id | manager_id | email | dept_id | dept_name |
+-----+-----+-----+-----+-----+-----+-----+
| 106 | Sneha | 55000.00 | NULL | 105 | sneha@gmail.com | NULL | NULL |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

```

```

mysql> SELECT *
-> FROM Departments
-> LEFT JOIN Employees
-> ON Departments.dept_id=Employees.dept_id;

```

```
mysql> SELECT *
-> FROM Departments
-> LEFT JOIN Employees
-> ON Departments.dept_id=Employees.dept_id;
```

dept_id	dept_name	emp_id	emp_name	salary	dept_id	manager_id	email
1	CSE	101	Rahul	55000.00	1	NULL	rahul@gmail.com
1	CSE	102	Priya	71500.00	1	101	priya@gmail.com
1	CSE	107	Kavya	68200.00	1	101	kavya@gmail.com
2	ECE	103	Amit	75000.00	2	101	amit@gmail.com
2	ECE	105	Rohan	90000.00	2	NULL	rohan@gmail.com
3	MBA	NULL	NULL	NULL	NULL	NULL	NULL
4	Civil	NULL	NULL	NULL	NULL	NULL	NULL

```
7 rows in set (0.00 sec)
```

```
mysql> SELECT
-> E.emp_name Employee,
-> M.emp_name Manager
-> FROM Employees E
-> LEFT JOIN Employees M
-> ON E.manager_id=M.emp_id;
```

Employee	Manager
Rahul	NULL
Priya	Rahul
Amit	Rahul
Rohan	NULL
Sneha	Rohan
Kavya	Rahul

```
6 rows in set (0.00 sec)
```

```
mysql> SELECT COUNT(*) AS TotalEmployees
-> FROM Employees;
```

TotalEmployees
6

```
mysql> SELECT COUNT(*) AS TotalEmployees
-> FROM Employees;
```

TotalEmployees
6

```
1 row in set (0.05 sec)
```

```
mysql> SELECT dept_id,
-> AVG(salary)
-> FROM Employees
-> GROUP BY dept_id;
```

dept_id	AVG(salary)
NULL	55000.000000
1	64900.000000
2	82500.000000

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT dept_id,
-> AVG(salary)
-> FROM Employees
-> GROUP BY dept_id
-> HAVING AVG(salary)>60000;
```

```
mysql> SELECT dept_id,
-> AVG(salary)
-> FROM Employees
-> GROUP BY dept_id
-> HAVING AVG(salary)>60000;
```

```
+-----+-----+
| dept_id | AVG(salary) |
+-----+-----+
|      1 | 64900.000000 |
|      2 | 82500.000000 |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT
-> MAX(salary),
-> MIN(salary)
-> FROM Employees;
```

```
+-----+-----+
| MAX(salary) | MIN(salary) |
+-----+-----+
|    90000.00 |    55000.00 |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM Employees
-> WHERE salary>
-> (
-> SELECT AVG(salary)
-> FROM Employees
-> );
```

```
+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | salary | dept_id | manager_id | email |
+-----+-----+-----+-----+-----+-----+
|    102 | Priya    | 71500.00 |      1 |      101 | priya@gmail.com |
|    103 | Amit     | 75000.00 |      2 |      101 | amit@gmail.com |
|    105 | Rohan    | 90000.00 |      2 |      NULL | rohan@gmail.com |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM Employees
-> WHERE dept_id=
-> (
-> SELECT dept_id
-> FROM Employees
-> WHERE emp_name='Rahul'
-> );
```

emp_id	emp_name	salary	dept_id	manager_id	email
101	Rahul	55000.00	1	NULL	rahul@gmail.com
102	Priya	71500.00	1	101	priya@gmail.com
107	Kavya	68200.00	1	101	kavya@gmail.com

3 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM Employees
-> WHERE salary=
-> (
-> SELECT MAX(salary)
-> FROM Employees
-> );
```

emp_id	emp_name	salary	dept_id	manager_id	email
105	Rohan	90000.00	2	NULL	rohan@gmail.com

1 row in set (0.00 sec)

```
mysql> SELECT dept_id
-> FROM Employees
-> GROUP BY dept_id
-> HAVING COUNT(*)>5;
Empty set (0.00 sec)
```

```
mysql> SELECT *
-> FROM
-> (
-> SELECT *,
-> DENSE_RANK() OVER(ORDER BY salary DESC) rnk
-> FROM Employees
-> )t
-> WHERE rnk=2;
```

emp_id	emp_name	salary	dept_id	manager_id	email	rnk
103	Amit	75000.00	2	101	amit@gmail.com	2

1 row in set (0.01 sec)

```
mysql> SELECT *
-> FROM Employees E
-> WHERE salary>
-> (
-> SELECT AVG(salary)
-> FROM Employees
-> WHERE dept_id=E.dept_id
-> );
```

emp_id	emp_name	salary	dept_id	manager_id	email
102	Priya	71500.00	1	101	priya@gmail.com
105	Rohan	90000.00	2	NULL	rohan@gmail.com
107	Kavya	68200.00	1	101	kavya@gmail.com

3 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM Customers
-> WHERE customer_id NOT IN
-> (
-> SELECT customer_id
-> FROM Orders
-> );
```

```

-> FROM Employees
-> WHERE dept_id=E.dept_id
-> );
+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | salary | dept_id | manager_id | email |
+-----+-----+-----+-----+-----+-----+
| 102 | Priya | 71500.00 | 1 | 101 | priya@gmail.com |
| 105 | Rohan | 90000.00 | 2 | NULL | rohan@gmail.com |
| 107 | Kavya | 68200.00 | 1 | 101 | kavya@gmail.com |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT *
-> FROM Customers
-> WHERE customer_id NOT IN
-> (
-> SELECT customer_id
-> FROM Orders
-> );
+-----+-----+
| customer_id | customer_name |
+-----+-----+
| 3 | Kiran |
+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT
-> P.product_name,
-> SUM(quantity) TotalSold
-> FROM Order_Details O
-> JOIN Products P
-> ON O.product_id=P.product_id
-> GROUP BY P.product_name
-> ORDER BY TotalSold DESC
-> LIMIT 1;
+-----+-----+
| product_name | TotalSold |
+-----+-----+
| Mouse | 8 |
+-----+-----+
1 row in set (0.00 sec)

```

```
mysql> SELECT dept_id,
-> COUNT(*) TotalEmployees
-> FROM Employees
-> GROUP BY dept_id
-> ORDER BY TotalEmployees DESC
-> LIMIT 1;
```

```
+-----+-----+
| dept_id | TotalEmployees |
+-----+-----+
|      1 |              3 |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT
-> MONTH(order_date) Month,
-> COUNT(*) Sales,
-> SUM(COUNT(*)) OVER(ORDER BY MONTH(order_date)) RunningTotal
-> FROM Orders
-> GROUP BY MONTH(order_date);
```

```
+-----+-----+-----+
| Month | Sales | RunningTotal |
+-----+-----+-----+
|      1 |      1 |              1 |
|      2 |      1 |              2 |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT
-> C.customer_name,
-> SUM(P.price*OD.quantity) TotalPurchase,
-> RANK() OVER(ORDER BY SUM(P.price*OD.quantity) DESC) CustomerRank
-> FROM Customers C
-> JOIN Orders O
-> ON C.customer_id=O.customer_id
-> JOIN Order_Details OD
-> ON O.order_id=OD.order_id
-> JOIN Products P
-> ON OD.product_id=P.product_id
-> GROUP BY C.customer_name;
```

```

-- GROUP BY C.customer_name,
+-----+-----+-----+
| customer_name | TotalPurchase | CustomerRank |
+-----+-----+-----+
| Ravi          | 121500.00     | 1            |
| Anita        | 4900.00       | 2            |
+-----+-----+-----+
2 rows in set (0.00 sec)

```

```

mysql> SELECT
  -> E.emp_name,
  -> E.salary,
  -> M.emp_name Manager,
  -> M.salary ManagerSalary
  -> FROM Employees E
  -> JOIN Employees M
  -> ON E.manager_id=M.emp_id
  -> WHERE E.salary>M.salary;
+-----+-----+-----+-----+
| emp_name | salary  | Manager | ManagerSalary |
+-----+-----+-----+-----+
| Priya    | 71500.00 | Rahul   | 55000.00      |
| Amit     | 75000.00 | Rahul   | 55000.00      |
| Kavya    | 68200.00 | Rahul   | 55000.00      |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

```

```

mysql> SELECT email,
  -> COUNT(*)
  -> FROM Employees
  -> GROUP BY email
  -> HAVING COUNT(*)>1;
Empty set (0.00 sec)

```